

Yun Tung, Chu

Last updated in Jan 2026

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Education

- MS in Electrical and Computer Engineering**, University of Washington, Seattle, WA Sep 2024 – Jun 2026
GPA: 3.9/4.0
- Relevant Coursework: Algorithms, Distributed Systems, AI For Mobile Robots, LLM, AR/VR
- BS in Management Information System**, National Chengchi University, Taipei, Taiwan Sep 2020 – Jan 2024
GPA: 4.04/4.3
- Relevant Coursework: Data Structure, Operating System, Database Management System

Work/Internship Experience

- AI/Software Engineering Intern — Innodisk Co.** Jul 2022 – Aug 2022
- Expanded image dataset by 4.5 times using gaussian data augmentation techniques, improving model training efficiency.
 - Optimized YOLOv4-tiny on Xilinx KV260 to detect defects, achieving 95% mAP and boosting inference by 50+ FPS.
 - Containerized project using Docker on Linux to reduce environment-related issues during deployment.
- Backend Engineering Intern — Hualiteq International Co.** Aug 2023 – Dec 2023
- Developed IVR backend logic using JavaScript/MSSQL and integrated Google TTS API for dynamic voice response, enabling real-time user interactions.
 - Implemented GitLab CI/CD practices across 10+ team members for IVR projects, enhancing development efficiency.

Relevant Experience

- Research Assistant — Information Processing Lab @University of Washington** Jul 2025 – Present
- Engineered automated synthetic data generation pipeline producing 50,000+ images, integrating Vega Prime rendering and GroundDINO (Open-Set Detection) for auto-labeling, reducing manual annotation time by 50%.
 - Optimized dataset composition to achieve 84% mAP on YOLOv8-tiny across five environmental conditions, meeting constraints for edge deployment.
 - Deployed SAM2 (Segment Anything Model 2) for zero-shot semantic segmentation of microscopic particles and engineered a geometric filtering algorithm using OpenCV to decouple boundaries from cracks.
 - Independently developed novel integration of Generative AI Foundation Models (Hunyuan3D-2) with depth-based layering in Blender to synthesize photorealistic 3D environments for training data.
- Distributed Systems: Sharded Key-Value Store (Course Project)** Sep 2025 – Dec 2025
- Architected a fault-tolerant, linearizable key-value storage system using Multi-Instance Paxos for consensus and log replication, ensuring data consistency across server failures and network partitions.
 - Implemented a fault-tolerant Shard Master service to manage dynamic re-sharding and load balancing, enabling replica groups to join/leave the cluster without service downtime.
 - Engineered cross-shard ACID transactions using Two-Phase Commit (2PC) with locking to ensure serializability and deadlock prevention in a distributed environment.
- Battery Data Standardized Web Solution — Full-stack Developer (Industrial Capstone)** Jan 2025 – Jun 2025
- Architected a modular data pipeline using FastAPI and S3-based storage; designed RESTful APIs with 15+ endpoints to handle multimodal data ingestion and real-time processing status.
 - Implemented frontend with React and secure authentication via OAuth2, providing clients with isolated access control and dynamic routing for project-specific data.
 - Leveraged OpenAI's API to intelligently match and standardize noisy column names across datasets, improving automation accuracy and reducing manual data mapping effort. Link: [Website](#)
- Color Vision Deficiency Assistance Software Module — Software Developer** Apr 2022 – Dec 2023
- 1st Place Winner, 2023 National Universities Innovation Competition: Built a MR app on Hololens2 for color-deficient users; integrated TensorFlow Object Detection (94% mAP) and optimized socket latency by 100ms. [Github Repo](#)

Technical Skills

Languages: Python, Java, JavaScript, PHP, HTML, SQL, C#, Object-oriented Programming

Tools & Frameworks: PyTorch, TensorFlow, Docker, Git, GitHub, AWS, FastAPI, React, SQL (MySQL/PostgreSQL), REST APIs